Last updated: April 2026

Hyungjin Kim

Korea Development Institute
263 Namsejong-ro, Sejong 30149, Republic of Korea
hyungjinkim@kdi.re.kr · hyungjinkim.com

Education

The Pennsylvania State University — May 2025
Ph.D. in Economics — *State College, PA*

Korea University — Feb 2019
M.A. in Economics — *Seoul, South Korea*

Korea University — Feb 2017
B.A. in Economics and Financial Engineering — *Seoul, South Korea*

Academic Positions

Korea Development Institute — July 2025 – Present
Associate Research Fellow — *Sejong, South Korea*

Fields of Interest

International Trade, Industrial Organization

Working Papers

The Coercive Turn Toward Swing States:
Vertical Bottlenecks and the Paradox of Competitiveness

Swing states—third-country firms sourcing critical inputs from China—are the target of US coercion, and as China grows more productive that coercion intensifies rather than relaxes. In a three-stage vertical game, bottleneck complementarity makes voluntary alignment too expensive for the US to buy through subsidies alone, so past a Chinese-productivity threshold policy switches from subsidies to subsidies plus coercion. An HS6 panel difference-in-differences recovers the predicted interaction in bottleneck sectors identified by the production-network literature, null or sign-flipped under non-bottleneck lists.

Export Controls and Endogenous Upstream Price Incidence:
The Chip War's Consequences for the AI-Chip Industry

Modern semiconductor production is vertical, granular, and concentrated. This paper shows that in a bilateral oligopoly, export controls force a renegotiation of upstream input prices — a channel absent from standard quantitative trade models that treat input costs as fixed. I develop a vertical production network model in which AI-chip designers negotiate with a concentrated set of High-Bandwidth Memory (HBM) suppliers via Nash-in-Nash bargaining. Using Korean customs data, I document destination-specific price divergences consistent with relationship-specific bargaining. In the calibrated model, targeted buyers' input prices fall while rivals' input prices rise — the upstream shock-absorber and cross-buyer spillover channels. The input-price adjustment is roughly one-fifth of the downstream markup response, and a frozen-input counterfactual that ignores this renegotiation overestimates the competitive reallocation caused by export controls. Network composition matters: adding a single qualified supplier dampens the targeted buyer's markup increase by approximately 46 percent.

When Do Standard Sector Taxonomies Fail? Welfare-Consistent Aggregation in Quantitative Trade Models
with Sungwan Hong

Quantitative trade analysis is inevitably an exercise in aggregation. We study when standard sector taxonomies are adequate for welfare analysis in multi-country trade models with production networks, and when welfare-consistent clustering does better. Starting from a high-resolution FIGARO-E3 benchmark, taxonomy-based partitions remain strong under mild compression and broad shocks, while welfare-consistent clustering does better when aggregation becomes coarse and shocks are upstream-intensive.

Trade, Multinational Production, and Regional Inequality
with Mira Rim

We study how China's reductions in trade and multinational production (MP) barriers reshaped regional inequality in Korea. A shift-share IV design shows that omitting the MP channel attenuates the estimated import-competition effect; a multi-country, multi-sector GE model with trade, MP, and internal migration quantifies the 2000–2007 China shock, yielding an aggregate welfare gain of $\sim$2.0% with per-worker gains of 1.7–2.5% across regions. A closed-form decomposition attributes 1.42 pp to trade, 0.55 pp to migration, and 0.04 pp to vertical MP. Jointly analyzing trade and MP is necessary both for causal identification and for accurate welfare accounting.

Triadic Gravity in Intermediated Petrochemical Trade
with Yang Shen

When firms can reroute shipments through intermediary hubs, the bilateral trade elasticity is strictly attenuated relative to the standard gravity benchmark — under Pareto productivity, by a factor equal to the share of trade flowing directly between origin and destination. Using transaction-level data from a Korean petrochemical trading house with 25 worldwide offices, we recover the structural attenuation parameter via a *Switch* control that absorbs mode selection. Standard gravity overstates route-specific trade-flow responses by 1.2–1.4$\times$ at cross-sector-average intermediary shares and 1.7–2.4$\times$ at shares plausible for petrochemical trade.

Equilibrium Uniqueness in Entry Games with Combinatorial Actions and Endogenous Pricing
with Sungwan Hong

We develop a uniqueness framework for combinatorial entry games with endogenous Atkeson–Burstein markups, decomposing the threshold-map Jacobian into density dampening, marginal-value sensitivity, and threshold location. When the resulting bathtub-shaped contraction modulus peaks below one, the Bayesian Nash equilibrium is unique and supermodularity is preserved so existing CDCP solvers remain applicable. In a quantitative application to global cloud infrastructure entry (five providers, 42 countries), the calibrated model matches 99.5% of in-sample entry decisions; endogenous markups sustain 15 additional firm-market pairs and lower consumer surplus by 10.6% relative to a constant-markup benchmark.

Export Controls, Directed Innovation, and Third-Party Incidence
with Sungwan Hong

Export controls do more than restrict trade: they change where the target innovates. In a three-player model, anticipated US controls redirect Chinese R&D toward restricted, hard-to-substitute inputs, shifting input-cost incidence onto exposed third-country firms that source Chinese intermediates. Chinese-assignee USPTO patenting rises in directly exposed chip codes around the May 2020 Huawei FDPR, and exposed East Asian firms facing larger China-origin input-cost shocks report higher input-cost shares; holding Chinese productivity fixed, total surplus slightly favors controls, but once China's R&D allocation responds the ranking flips toward credible restraint.

Publications

Kim, Hyungjin, and Jinhyuk Lee. "Collection of personal data with information externality in two-sided markets." *Journal of Economic Theory and Econometrics* 31.3 (2020): 1–22. *(pre-PhD)*

Research Experience

Research Assistant for Nonna Sorokina 2024–2025
The Pennsylvania State University

Research Assistant for Jonathan Eaton 2022
The Pennsylvania State University

Research Assistant for Seunghan Yoo 2017–2019
Korea University

Teaching Experience

Instructor of Record Summer 2021
The Pennsylvania State University

- ECON 302 Intermediate Microeconomic Analysis

Teaching Assistant 2020–2024
The Pennsylvania State University

- ECON 470 International Trade and Finance (SP24)
- ECON 434 International Finance (FA20, SP21, FA21, FA22, SP23)
- ECON 437 Multinationals and the Globalization of Production (SP22)
- ECON 333 International Trade (SU20)
- ECON 444 Economics of Corporation (SP20)

Invited Seminars

2026 University of Pittsburgh; Korea Academic Society of Industrial Organization (Seoul)

2025 Korea Development Institute; Bank of Korea; Korea University; Hanyang University; Korea Institute of Finance; Korea Insurance Research Institute; Korea Capital Market Institute; Korea Institute for International Economic Policy; Korea Institute for Industrial Economics and Trade; Korea Information Society Development Institute; Samsung Economic Research Institute; Center for International Development

Conference Presentations

2026 Midwest International Trade Conference (Ohio State); IMF Economic Review (Bangkok); Sapporo Workshop on Economic Theory (*scheduled*)

2025 Econometric Society World Congress (Seoul)

2024 American Economic Association annual meeting (San Antonio); Economics Graduate Student Conference (Washington University in St. Louis); Comparative Analysis of Enterprise Data (Penn State)

2023 International Economic Association World Congress (Medellin); Midwest International Trade (Georgia Tech); International Trade and Finance Association (Richmond); International Atlantic Economic Society (Philadelphia); CIREQ Ph.D. Conference (Université de Montréal)

Awards & Honors

Fellow Paper, 33rd Annual Conference
International Trade and Finance Association — *2023*

Doctoral Study Abroad Scholarship
Yongwoon Foundation — *2019*

Brain Korea 21 Plus Scholarship
Korea University — *2017–2019*

Specialized Skills

Programming Languages: MATLAB, STATA
Language: English (fluent), Korean (native), Japanese (beginner)

Nationality

Citizenship: South Korea

References

Prof. Stephen Yeaple (Chair)
Professor of Economics
The Pennsylvania State University
+1 (814) 865-5452
sry3@psu.edu

Prof. James R. Tybout
Professor of Economics
The Pennsylvania State University
+1 (814) 865-4259
jxt32@psu.edu

Prof. Jingting Fan
Associate Professor of Economics
Cheung Kong Graduate School of Business
jtfan@ckgsb.edu.cn